

Insurance Policy & Claim Management System

System Testing, Test Execution Reports & Comprehensive API Verification Results

TEST ENVIRONMENT

Spring Boot 3.3.3 · PostgreSQL 16 · Java 17

AUTOMATED TESTS

23 Tests (100% Passed)

API ENDPOINTS

22 Endpoints Verified

DOCUMENT DATE

September 2026

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SECTION 01

Executive Metrics & Health Overview

23 / 23

BACKEND TESTS PASSED

100%

AUTOMATED PASS RATE

22

API ENDPOINTS TESTED

0

BUILD / LINTER ERRORS

All test scenarios were executed against live instances of the Spring Boot 3 API service and PostgreSQL 16 database. Results validate full compliance with Software Requirements Specification (SRS) functional requirements FR1 through FR7.



SECTION 02

Automated Backend Test Suite Execution

Automated regression tests run via Apache Maven (`mvn test`) across all application layers:

Test Suite Class	Tests	Failures	Errors	Time	Status
BackendApplicationTests	1	0	0	11.41s	PASSED
JwtTokenProviderTest	2	0	0	0.40s	PASSED
AmlServiceTest	4	0	0	1.65s	PASSED
AuditLogServiceTest	2	0	0	0.24s	PASSED
ClaimServiceTest	4	0	0	0.28s	PASSED
PolicyEndorsementTest	4	0	0	0.15s	PASSED
UnderwriteServiceTest	4	0	0	0.10s	PASSED
UserServiceTest	2	0	0	0.35s	PASSED



SECTION 03

Authentication API Endpoint Tests

POST	/api/auth/login	200 OK - Administrator Authentication
REQUEST PAYLOAD		CONTENT-TYPE: APPLICATION/JSON
<pre>{ "email": "admin@ipcms.local", "password": "password123" }</pre>		
RESPONSE PAYLOAD		STATUS: 200 OK
<pre>{ "token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9 ... ", "type": "Bearer", "email": "admin@ipcms.local", "username": "admin", "role": "ADMIN" }</pre>		

POST**/api/auth/login (Negative Test)****401 Unauthorized** - Invalid Password**REQUEST PAYLOAD****CONTENT-TYPE: APPLICATION/JSON**

```
{
  "email": "admin@ipcms.local",
  "password": "wrongpassword"
}
```

RESPONSE PAYLOAD**STATUS: 401 UNAUTHORIZED**

```
{
  "timestamp": "2026-09-10T08:42:10.112",
  "status": 401,
  "error": "Unauthorized",
  "message": "Bad credentials"
}
```

**SECTION 04****Policies & Onboarding Endpoint Tests (FR1, FR4, FR5)**

POST**/api/policies/onboard****200 OK** - Policyholder Self-Registration (FR1)**REQUEST PAYLOAD****APPLICANT DATA**

```
{
  "username": "davidmiller",
  "email": "david.miller@ipcms.local",
  "password": "password123",
  "legalName": "David Miller",
  "taxId": "TAX-USA-98765",
  "address": "742 Evergreen Terrace, Springfield",
  "contactNumber": "+1-555-8392"
}
```

RESPONSE PAYLOAD**STATUS: 200 OK**

```
{
  "id": 6,
  "username": "davidmiller",
  "email": "david.miller@ipcms.local",
  "role": "POLICYHOLDER",
  "amlStatus": "PENDING",
  "kycStatus": "VERIFIED",
  "message": "Registration submitted. Asynchronous AML verification in progress."
}
```

POST**/api/policies/1/endorse****200 OK**

- Policy Endorsement & Pro-Rata Math (FR4)

REQUEST PAYLOAD**ENDORSEMENT DELTA**

```
{
  "productType": "Commercial Cargo (Hazard Tier 2)",
  "basePremium": 19500.00
}
```

RESPONSE PAYLOAD**STATUS: 200 OK**

```
{
  "policyNumber": "POL-2024-001",
  "oldPremium": 14500.00,
  "newPremium": 19500.00,
  "proRataAdjustment": 616.44,
  "daysRemaining": 45,
  "message": "Policy endorsed successfully. Pro-rata premium adjustment calculated."
}
```

GET**/api/policies/1/history****200 OK**

- Immutable Historical Snapshots (FR4, FR6)

RESPONSE PAYLOAD**ARCHIVED BASELINES**

```
[
  {
    "id": 1,
    "policyNumber": "POL-2024-001",
    "productType": "Commercial Cargo",
    "basePremium": 14500.00,
    "activeReserve": 0.00,
    "expiryDate": "2026-10-25",
    "status": "Active",
    "modifiedAt": "2026-09-10T08:52:14.201",
    "modifiedBy": "underwriter@ipcms.local"
  }
]
```

**SECTION 05****Claims Management & FNOL Tests (FR3)**

POST**/api/claims/fnol****201 Created**

- FNOL Registration within 14 Days

REQUEST PAYLOAD**INCIDENT REPORT**

```
{
  "policyNumber": "POL-2024-001",
  "incidentDate": "2026-09-08",
  "incidentType": "Collision",
  "incidentLocation": "Port of Rotterdam",
  "lossDescription": "Docking vessel collision with cargo crane during squall",
  "initialEstimate": 8500.00
}
```

RESPONSE PAYLOAD**STATUS: 201 CREATED**

```
{
  "id": 1,
  "claimNumber": "CLM-2026-0001",
  "policyNumber": "POL-2024-001",
  "policyholderName": "John Meridian",
  "incidentType": "Collision",
  "incidentDate": "2026-09-08",
  "grossReserve": 0.00,
  "lossDescription": "Docking vessel collision with cargo crane during squall",
  "status": "Pending FNOL"
}
```

POST**/api/claims/fnol (Cutoff Validation Test)****400 Bad Request**

- Incident Over 14 Days

REQUEST PAYLOAD (INCIDENT 26 DAYS OLD)**VALIDATION FAILURE**

```
{
  "policyNumber": "POL-2024-001",
  "incidentDate": "2026-08-15",
  "lossDescription": "Old damage report"
}
```

RESPONSE PAYLOAD**STATUS: 400 BAD REQUEST**

```
{
  "status": 400,
  "error": "Bad Request",
  "message": "Incident date exceeds allowable FNOL notification window (maximum 14 days)"
}
```

PUT**/api/claims/1/reserve****200 OK**

- Reserve Adjustment & Auto-Transition

REQUEST PAYLOAD**GROSS RESERVE VALUE**

```
{
  "grossReserve": 16500.00
}
```

RESPONSE PAYLOAD**STATUS: 200 OK**

```
{
  "id": 1,
  "claimNumber": "CLM-2026-0001",
  "grossReserve": 16500.00,
  "adjuster": "Stefan Lindqvist",
  "status": "Under Investigation"
}
```

**SECTION 06****Underwriting Matrix & Risk Scoring Tests (FR2)****POST****/api/underwrite/quote****200 OK**

- Premium Quote Calculation

REQUEST PAYLOAD (RISK FACTOR SLIDERS)**VALUES: 0-100**

```
{
  "regionFactor": 75,
  "assetAgeFactor": 45,
  "priorClaimsFactor": 15,
  "businessTypeFactor": 65
}
```

RESPONSE PAYLOAD**FORMULA: \$500 + (\$15 * AVG)**

```
{
  "compositeRiskScore": 50,
  "riskLevel": "MEDIUM RISK",
  "premiumQuote": 1250.00
}
```


POST**/api/underwrite/overrides/1/signoff****200 OK** - Supervisor Override Approval**REQUEST PAYLOAD****SUPERVISOR SIGN-OFF**

```
{
  "approved": true
}
```

RESPONSE PAYLOAD**STATUS: 200 OK**

```
{
  "id": 1,
  "policyNumber": "POL-2024-001",
  "overrideType": "Premium Rate Reduction",
  "deltaValue": "-12%",
  "requestedBy": "underwriter@ipcms.local",
  "status": "APPROVED",
  "supervisorSignoffBy": "admin@ipcms.local"
}
```

**SECTION 07****Administration & Audit Trail Tests (FR6)**

GET

/api/admin/audit-logs

200 OK

- Chronological Audit Trail

RESPONSE PAYLOAD

AUDIT RECORDS

```
[
  {
    "id": 4,
    "timestamp": "2026-09-10T08:53:55.302",
    "userEmail": "admin@ipcms.local",
    "action": "OVERRIDE_APPROVED",
    "target": "POL-2024-001",
    "detail": "Supervisor signed off on Premium Rate Reduction (-12%)"
  },
  {
    "id": 3,
    "timestamp": "2026-09-10T08:53:15.890",
    "userEmail": "claims_handler@ipcms.local",
    "action": "RESERVE_UPDATED",
    "target": "CLM-2026-0001",
    "detail": "Gross reserve revised to $16,500.00"
  }
]
```

SECTION 08



Requirements Traceability Matrix (RTM)

Req ID	Functional Requirement	Tested Endpoints	Coverage	Status
FR1	Policyholder Onboarding	POST /api/policies/onboard	100%	VERIFIED
FR2	Dynamic Underwriting Matrix	POST /api/underwrite/quote, POST /api/underwrite/overrides	100%	VERIFIED
FR3	Claims Registration & FNOL	POST /api/claims/fnol, PUT /api/claims/{id}/reserve	100%	VERIFIED

Req ID	Functional Requirement	Tested Endpoints	Coverage	Status
FR4	Policy Endorsements & Snapshots	POST /api/policies/{id}/endorse, GET /api/policies/{id}/history	100%	VERIFIED
FR5	Asynchronous AML Watchlist Check	POST /api/policies/onboard (OFAC Match)	100%	VERIFIED
FR6	Audit Trail & Governance	GET /api/admin/audit-logs	100%	VERIFIED
FR7	Authentication & User Registry	POST /api/auth/login, PUT /api/admin/users/{id}/toggle-active	100%	VERIFIED